

CHAPTER 2, VALUE OF MONEY: MEANING AND THEORIES OF VALUE OF MONEY

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The Quantity Theory of Money

A simple theory linking the inflation rate to the growth rate of the money supply.

Begins with the concept of **velocity** ...

Velocity

- basic concept: the rate at which money circulates
- definition: the number of times the average dollar bill changes hands in a given time period
- example: In 2007,
 - \$500 billion in transactions
 - money supply = \$100 billion
 - The average dollar is used in five transactions in 2007
 - So, velocity = 5

Velocity

- This suggests the following definition:
where

$$V = \frac{T}{M}$$

V = velocity

T = value of all transactions

M = money supply

Velocity

Use nominal GDP as a proxy for total transactions.

Then,

Where

$$V = \frac{P \times Y}{M}$$

P = price of output (GDP deflator)

Y = quantity of output (real GDP)

P* × *Y = value of output (nominal GDP)



The quantity equation

- The **quantity equation**

$$M \times V = P \times Y$$

follows from the preceding definition of velocity.

- It is an *identity*:
it holds by definition of the variables.



Money demand and the quantity equation

- M/P = **real money balances**, the purchasing power of the money supply.

- A simple money demand function:

$$(M/P)_d = kY$$

where

k = how much money people wish to hold for each dollar of income.

(k is exogenous)



Money demand and the quantity equation

- money demand: $(M/P)_d = kY$
- quantity equation: $M \times V = P \times Y$
- The connection between them: $k = 1/V$
- When people hold lots of money relative to their incomes (k is high), money changes hands infrequently (V is low).



Back to the quantity theory of money

- starts with quantity equation
- assumes V is constant & exogenous: $V = \bar{V}$
- With this assumption, the quantity equation can be written as

$$\overline{MV} = PY$$



The quantity theory of money, *cont.*

$$\overline{M} = V \times P \times Y$$

How the price level is determined:

- With **V** constant, the money supply determines nominal GDP (**P** × **Y**).
- Real GDP is determined by the economy's supplies of **K** and **L** and the production function.
- The price level is
 $P = (\text{nominal GDP})/(\text{real GDP}).$

What is Money?

money

Definition: Money. The Economists Way.

Money is anything that is generally accepted in payment for goods or services or in the repayment of debts.

Then, what is the thing in your left pocket? **Currency** is the bills and coins of a country.

Is money a **wealth**?

Definition: Personal wealth

Wealth is the *stock* of money but also other assets such as bonds, common stock, art, land, furniture, cars, and houses, taken together.

Definition: Income

Income is the *flow* of money, interest rate and maturity payments per unit of time.

The connection between money and prices

- Inflation rate = the percentage increase in the average level of prices.
- Price = amount of money required to buy a good.
- Because prices are defined in terms of money, we need to consider the nature of money, the supply of money, and how it is controlled.

What are the functions of money?

Functions of money:

- **A medium of exchange:** it is used to pay for goods and services
 - imagine a barter economy
 - facilitating transactions (reducing transaction costs) encouraging
 - specialization
- **A unit of account:** to measure the value of the goods and services
 - imagine a barter economy, 10 goods only price normalization
- **A store of value:** it has most of its purchasing power tomorrow too
 - any asset is a store of value. Why people choose money? liquidity
 -
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What is liquidity?

Definition: Liquidity

Liquidity is the relative ease and speed with which an asset can be converted into a medium of exchange

Advantages and disadvantages of a higher liquidity:

Advantages

- lower costs of transactions
- ...

Disadvantages

- no interest inflation eats it
- up
(hyperinflation in BG, 1997)

How do we measure money?

Why is measuring money important?

Measures of the money (money supply): **monetary aggregates**

Definition: Monetary Aggregates

Monetary Aggregates are the various measures of money supply used by the central banks around the world

Three main monetary aggregates: M1

- M2 M3



What is **M1**?: Currency and assets that can be used directly as a medium of exchange

Table 1 Measures of the Monetary Aggregates

	Value as of December 2002 (\$billions)
M1 = Currency	626.5
+ Traveler's checks	7.7
+ Demand deposits	290.7
+ Other checkable deposits	281.2
Total M1	1,206.1

M2 and M3

What is **M2**?: Currency and assets that are very liquid - can be easily turned into cash but are not cash themselves. M2 are the very short-term money market instruments

What is **M3**?: Currency and assets that are less liquid - can not be turned into cash with little cost

M2 = M1

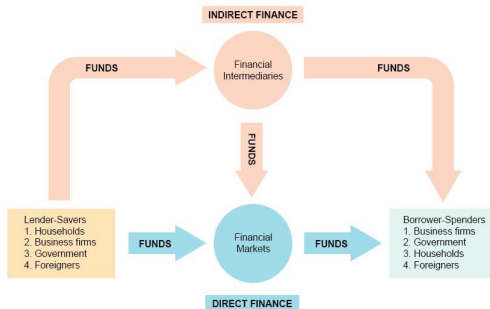
+ Small-denomination time deposits and repurchase agreements	1,332.3
+ Savings deposits and money market deposit accounts	2,340.4
+ Money market mutual fund shares (noninstitutional)	923.7
Total M2	5,802.5

M3 = M2

+ Large-denomination time deposits and repurchase agreements	1,105.2
+ Money market mutual fund shares (institutional)	767.7
+ Repurchase agreements	511.7
+ Eurodollars	341.1
Total M3	8,528.2

The function of financial markets

The main function of financial markets is to channel funds. How? From whom? To whom?



In *direct finance*, borrowers borrow funds directly from lenders in financial markets by selling them securities.

In *indirect finance*, borrowers borrow funds from the financial intermediaries.

The importance of financial markets

Why is the channeling function important?

- Markets create value: savers get interest rate that they can spend after
- Markets create value: investors make profits over and above the interest payments that they can reinvest
-

Debt and Equity Markets

A firm or an individual can obtain funds in a financial market in two ways:

- issue a debt instrument (a bond)

Definition: A bond and its Maturity

A bond is a debt security that promises to make payments periodically for a specified period of time until a specified date.

The maturity of a debt instrument is the number of years until that instrument's expiration date.

- issue equities, such as stocks

Definition: A stock (equity)

Stocks (equities) are claims to a share in the net income (income after expenses and taxes) and the assets of a business

Term Structure of the Debt and Equity

A debt instrument is:

- short-term, if its maturity is less than a year long-term, if its maturity
- is ten years or longer
- intermediate-term, if its maturity is between 1 and 10 years

Equity instruments are only long-term! Why? They do not have a maturity.

Primary and Secondary Markets

What is a primary market?

Definition: Primary Markets

A primary market is a financial market in which new issues of a security, such as a bond or a stock, are sold to initial buyers by the corporation or government agency who wants to borrow funds.

What is a secondary market?

Definition: Secondary Markets

A secondary market is a financial market in which securities that have been previously issued can be resold.

Primary and Secondary Markets

Things to remember about primary and secondary markets:

- A firm acquires new funds only when its securities are first sold in the primary market
- Once the security is sold on the primary market, those who trade it on the secondary markets win or lose money
- However, for whom do you think the secondary market is most relevant?

Organization of a Secondary Market

Secondary markets are organized in two ways:

- **Exchanges:** where buyers and sellers of securities meet in one central location to conduct trades. Examples?
- **Over-the-counter markets:** dealers at different locations who have securities and stand ready to sell them. They are usually computer-traded.

What is the difference between money and capital markets?

- **Money markets:** a market where short-term securities are traded (maturity less than 1 year)
- **Capital markets:** a market where long-term securities are traded - bonds and equity with maturity of more than 1 year

Financial Market Instruments

Instruments on the Money Markets

What are the instruments on the money market:

- **United States Treasury Bills:** U.S. government debt issued in 3-, 6-, and 12-month maturities to finance the federal government
 - pay no interest but sold at a discount
 - least risky bonds due to very low risk of *default*. Why?
- **Certificates of Deposit (CODs):** a debt instrument, sold by a bank to its depositors, that pays annual interest of a given amount and at maturity, pays back the original purchase price
- **Commercial Papers:** Commercial paper is a short-term debt instrument issued by large banks and big corporations, such as General Motors and AT&T that serves as a substitute for issuing bonds or taking loans
- **Repurchase Agreements (REPOs):** short-term loans (usually with a maturity of less than two weeks) in which Treasury bills serve as *collateral*
- **Federal (Fed) Funds:** overnight loans between banks

Financial Market Instruments

Instruments on the Capital Markets

- **Corporate stocks:** claims on the net income and assets of the corporations
 - by far the largest component on the capital market
 - considered a risky instrument for investors
- **Mortgages:** loans to households or firms to purchase housing, land, or other real structures, where the structure or land itself serves as collateral for the loans
- **Mortgage-Backed Securities:** securities in which mortgages are bundled and sold on the secondary market
- **Corporate Bonds:** long-term bonds issued typically by large corporations paying interest twice per year, and paying off face value at maturity date
- **Convertible corporate bonds:** bonds with the option of converting into a stock any time before maturity
- **U.S. Government Securities:** long-term debt of the US treasury to finance government deficits

Financial Intermediaries

Three large types of financial intermediaries:

- Depository institutions (banks): accept deposits, make loans

- Commercial banks
- Saving and loan associations (S&Ls) Credit unions

Contractual saving institutions

- Pension funds Insurance companies

Investment intermediaries Finance companies Mutual funds

- Money market mutual funds



Aggregate Output

Gross Domestic Product (GDP) = Value of all final goods and services produced in domestic economy during year

Aggregate Income

Total income of factors of production (land, capital, labor) during year

Distinction Between Nominal and Real

Nominal = values measured using current prices

Real = quantities, measured with constant prices

Aggregate Price Level

$$\text{GDP Deflator} = \frac{\text{nominal GDP}}{\text{real GDP}}$$

$$\text{GDP Deflator} = \frac{\$10 \text{ trillion}}{\$9 \text{ trillion}} = 1.11$$

Consumer Price Index (CPI) price of “basket” of goods and services



Growth Rates and the Inflation Rate

$$\text{Growth Rate} = \frac{X_2 - X_1}{X_1} \times 100$$

$$\text{GDP Growth Rate} = \frac{\$9.5 \text{ trillion} - \$9 \text{ trillion}}{\$9 \text{ trillion}} \times 100 = 5.6\%$$

$$\text{Inflation Rate} = \frac{113111 - 111111}{111111} \times 100 = 1.8\%$$

